Quiz on Wage Functions

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Question 1

Consider a one-period matching model in which wages are determined by Nash bargaining between workers and firms. That is, the wage W maximizes $(a - W)^{1-\beta} \times (W - z)^{\beta}$, where a is labor productivity, z is the utility from nonwork, and $\beta \in [0, 1]$ is workers' bargaining power. Then, the wage is given by:

A) $W = (1 - \beta) \times a + \beta \times z$

B) $W = \beta \times a + (1 - \beta) \times z$

C) $W = \beta \times (a + z)$

D) $W = (1 - \beta) \times (a + z)$

E) $W = \beta$

F) $W = 1 - \beta$

G) None of the above

Question 2

Consider a matching model in which firms set wages by sharing surplus with workers. We expect wages to be higher when:

A) Labor market tightness is lower.

B) Labor market tightness is higher.

C) Unemployment insurance is less generous.

D) Unemployment insurance is more generous.

E) Workers have less bargaining power.

F) Workers have more bargaining power.

G) None of the above.

Question 3

In the United States, what is a plausible estimate of the elasticity γ of real wages with respect to productivity?

A) $\gamma = 0$

B) $\gamma = 0.1$

C) $\gamma = 0.5$

D) $\gamma = 0.9$

E) $\gamma = 1$

F) None of the above

Question 4

The surplus enjoyed by a worker from a worker-firm match is $(W - z)/[s + f(\theta)]$, where W is the wage, z is the value from unemployment, s is the job-separation rate, and $f(\theta)$ is the job-finding rate. Why is the term $s + f(\theta)$ in the denominator of the surplus?

A) Because $s + f(\theta)$ is the expected duration of unemployment for a worker who just lost her job.

B) Because $s + f(\theta)$ is the expected duration of employment for a worker who just found a job.

C) Because $1/[s + f(\theta)]$ is the expected duration of unemployment for a worker who just lost her job.

D) Because $1/[s + f(\theta)]$ is the expected duration of employment for a worker who just found a job.

E) Because $s + f(\theta)$ is the expected duration of the period during which a worker initially employed and a worker initially unemployed retain a different employment status.

F) Because $1/[s + f(\theta)]$ is the expected duration of the period during which a worker initially employed and a worker initially unemployed retain a different employment status.

G) None of the above.

Question 5

Consider a matching model with a fixed wage. In the tightness-employment diagram, an increase in the wage leads to:

A) An inward shift of the labor supply curve

B) An outward shift of the labor supply curve

C) A downward shift of the labor demand curve

D) An upward shift of the labor demand curve

E) A downward rotation of the labor demand curve

F) An upward rotation of the labor demand curve

G) None of the above

Question 6

Consider a matching model with Nash bargaining. Imagine that the government implements training programs to increase the skills and productivity of workers. In the tightness-employment diagram, this policy would:

A) Raise the real wage

B) Lower the real wage

C) Shift the labor supply curve leftward

D) Shift the labor supply curve rightward

E) Have no effect on real wage and labor supply